HDOC Day-44 Activity

Menu-Driven C Program - Reverse Each Word and Print Name Initials

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Aim

To write a menu-driven C program using separate functions to reverse every word of a sentence without changing the word order and to print the initials of a name.

Operations

1. Reverse each word in a sentence without changing the order of words.
2. Print the initials of a name.
3. Exit.

Algorithm

Reverse each word:

1. Read the complete sentence using fgets().
2. Move through the sentence word by word.
3. Find the starting and ending position of each word.
4. Swap characters from both ends of that word until the middle is reached.
5. Keep spaces and word positions unchanged.

Print initials:

1. Read the complete name.
2. Print the first alphabetic character as an initial.
3. Whenever a space is followed by a letter, print that letter as the next initial.
4. Convert initials to uppercase and place a dot after each one.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void reverseRange(char str[], int start, int end)
{
    char temp;

    while (start < end)
    {
        temp = str[start];
        str[start] = str[end];
        str[end] = temp;
        start++;
        end--;
    }
}

void reverseEachWord(char sentence[])
{
    int i = 0, start;
```

```
    while (sentence[i] != '\0')
    {
        while (sentence[i] == ' ')
            i++;

        if (sentence[i] == '\0' || sentence[i] == '\n')
            break;

        start = i;

        while (sentence[i] != ' ' &&
               sentence[i] != '\0' &&
               sentence[i] != '\n')
        {
            i++;
        }

        reverseRange(sentence, start, i - 1);
    }
}

void printInitials(const char name[])
{
    int i;
    int newWord = 1;

    printf("Initials: ");

    for (i = 0; name[i] != '\0'; i++)
    {
        if (isspace((unsigned char)name[i]))
        {
            newWord = 1;
        }
        else if (newWord && isalpha((unsigned char)name[i]))
        {
            printf("%c.", toupper((unsigned char)name[i]));
            newWord = 0;
        }
        else
        {
            newWord = 0;
        }
    }

    printf("\n");
}

int main()
{
    int choice;
    char sentence[200];
    char name[150];

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(sentence, sizeof(sentence), stdin);

                reverseEachWord(sentence);

                printf("After reversing each word: %s", sentence);
                break;

            case 2:
                printf("Enter a name: ");
```

```
                fgets(name, sizeof(name), stdin);

                printInitials(name);
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Sample Output

```
----- MENU -----
1. Reverse each word in a sentence
2. Print initials of a name
3. Exit
Enter your choice: 1
Enter a sentence: What is special?
After reversing each word: tahW si ?laiceps

Enter your choice: 2
Enter a name: Alexander Graham Bell
Initials: A.G.B.

Enter your choice: 2
Enter a name: Gurveer Singh Cheema
Initials: G.S.C.

Enter your choice: 3
Exiting program...
```

Function Explanation

reverseRange(): Reverses characters between two positions by swapping the first and last characters, then moving toward the center.

reverseEachWord(): Locates every word separately and calls reverseRange() for that word. Therefore, the order of words remains unchanged.

printInitials(): Detects the first character of every word in the name, converts it to uppercase using toupper(), and prints a dot after it.

Easy Memory Tricks

Reverse a word: Start + End -> Swap -> Move inward.
Initials: First letter + letter after each space.
Example: Gurveer Singh Cheema -> G.S.C.

Important Note

The sample in the activity shows the name 'Alexander Grahambell' producing A.G. If the input contains three separate words such as 'Alexander Graham Bell', the program correctly prints A.G.B.

Result

Thus, the menu-driven C program was successfully implemented using separate functions to reverse each word of a sentence without changing word order and to display the initials of a name.

Quick Viva Questions

1. Why is fgets() used? It reads a complete line including spaces.
2. Does reverseEachWord() reverse the entire sentence? No. It reverses each word individually while preserving word order.
3. Which function converts a letter to uppercase? toupper().
4. Which header contains toupper(), isalpha(), and isspace()? ctype.h.
5. What are the initials of Gurveer Singh Cheema? G.S.C.
6. What is the output for 'What is special?'? tahW si ?laiceps.